

Module 6

classmate

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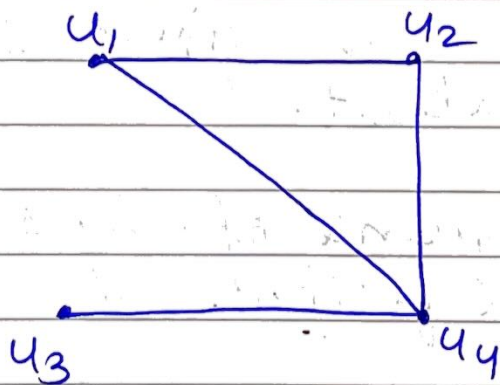
Case Study 3 Facebook Friend Recommendation using Graph Mining

V1: Link prediction problem:-

Graph has edges which connect each nodes/vertex.

V2: Overview of Graphs: node/vertex, edge/link, directed-edge, path.

Graph is represented using a set of vertices/nodes.

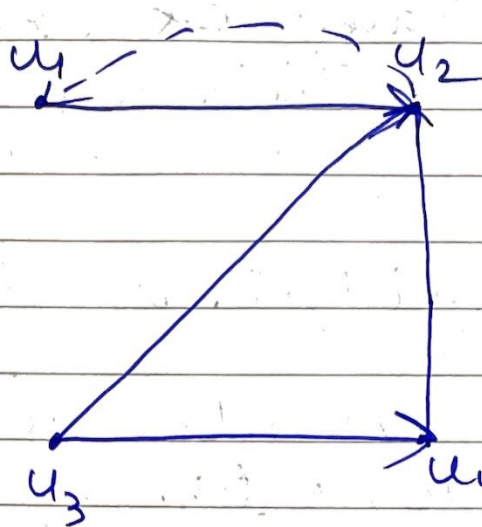


u_1, u_2 has edge b/w them.

undirected edges in the graph

$G = \langle V, E \rangle$ set of vertices & set of edges is a Graph.

edge = (u_i, u_j)



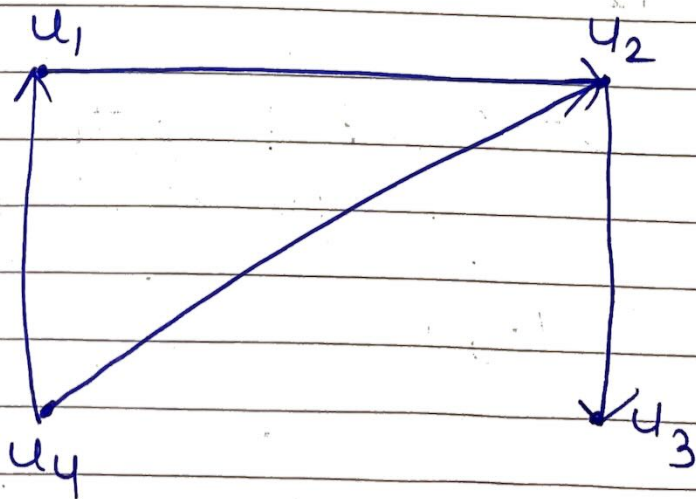
Directed edges

u_1 follows u_2 .

But u_2 also could follow u_1 .

So we are given directed edges dataset.

Path :- Sequence of edges that you can follow.



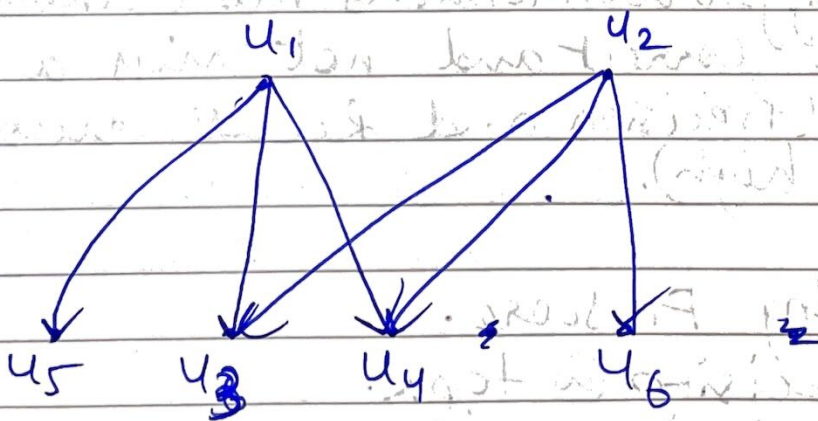
Path = $u_1 \rightarrow u_2 \rightarrow u_3$ (length = 2) (Since there is no edge b/w u_1 & u_3)

$\sim$ ly, $u_1 \rightarrow u_2 \rightarrow u_3$ (length = 3)

We are given a directed graph. It might be from Instagram.

v4: If there is an edge b/w 2 vertices, we label them as 1.

If there is no edge b/w 2 vertices, we label them as 0.



$u_1 \rightarrow \{u_3, u_4, u_5\}$

$u_2 \rightarrow \{u_3, u_4, u_6\}$

If u_1 follows u_2 we label it as 1.

Chance/Prob that u_2 follows u_1 is very high.

Graph Features :-

- 1) # of common vertices being followed by both u_1 and u_2
- 2) is there an edge $u_2 \rightarrow u_1$.

V5: Business Constraints and Metrics

- 1) No latency requirement.
- 2) Predict probability of a link.
- 3) Suggest connections most likely to be correct and not miss any.
(Precision and Recall must be high).

Using F1 score.

Precision @ topk.

Confusion matrix

V6: EDA: Basic Stats

networkx library is useful in graphs

V7: EDA: Follower and following state

Percentile $\rightarrow$ 90%

V8: EDA: Binary Classification Task

V9: Train and Test Split: EDA

Data changes temporarily (with time)

So we need to change our data according to time.

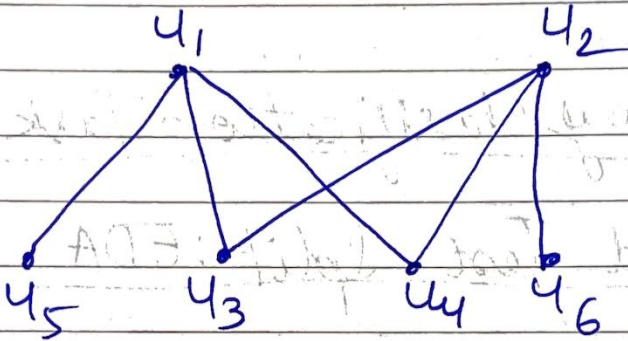
So ideally we need to do time based split. But we do not have time stamps with us.

So we will stick with Random Split only.

80:20 train test split.

V10: Feature Engineering on Graphs: Jaccard and Cosine Similarities

$$\text{Jaccard distance} = \frac{|x \cap y|}{|x \cup y|}$$



J. distance $\Rightarrow$ b/w u_1 & u_2
 of
 followers =
 of u_1 & u_2

$$|u_1 \cap u_2|_{\text{followers}} = u_3, u_4 = 2.$$

$$|u_1 \cup u_2| = u_3, u_4, u_5, u_6.$$

$$= \frac{2}{4} = \frac{1}{2}$$

Cosine distance b/w 2 vectors.

Otsuka - Ochiai Coefficient:-

It can be calculated b/w 2 sets.

$$\frac{|x \cap y|}{\sqrt{|x| \cdot |y|}}$$

VII: PageRank

Named after Larry Page of Google.

Internet is all about pages.

If there is a hyperlink on a webpage, the web is a structure of directed graph.

A webpage has larger value if it has a link from other webpages. If those pages are also very important then that webpage which is being linked to also gets increased.

PageRank gives the probability value for each of the webpage. This prob. is prob value to which a webpage lands into.

So, it gives a score to a vertex on a directed graph.

Networkx already has pagerank implemented.

We can also use log probability instead of simple probabilities.

V12: Shortest Path

V13: Connected Components

Strongly Connected Components: is a subgraph where there is a path from every node to every other node.

Weakly Connected Components: is a subgraph where there is a path from every node to every other node provided we ignore the directions.

Weakly Connected Components form groups or community in Social Network.

We can use networkx to implement

Weakly connected Component.

V14: Adar Index

~~Neighbourhood of point which is in set of points which are intersection of Neighbourhood of x and Neighbourhood of y .~~
 ~~$N(u)$ = Neighbourhood of u .~~

Adar Index $(x, y) =$

$$\sum_{u \in N(x) \cap N(y)} \left\{ \frac{1}{\log(|N(u)|)} \right\}$$

$N(u)$ = Neighbourhood of point u (which is in set of points which are intersection of Neighbourhood of x and Neighbourhood of y .)

Also can be said:-

Common elements with very large neighbourhoods are lesser significant when predicting a connection between two nodes

compared with elements shared between a small number of nodes.

V15: Katz Centrality

Similar to Pagerank. It is a feature will be used for graph problems. Available in networkx

V16: HITS Score

Available in networkx.

HITS—Hyperlink-Induced Topic Search. Also known as hubs & authorities. HITS gives Score $\langle h, a \rangle$.

Initialization condition $\forall$ page p
 $auth(p) = 1$ and $hub(p) = 1$.

Authority update rule :-

$$auth(p) = \sum_{i=1}^n hub(i)$$

Hub update rule :-

$$hub(p) = \sum_{i=1}^n auth(i)$$

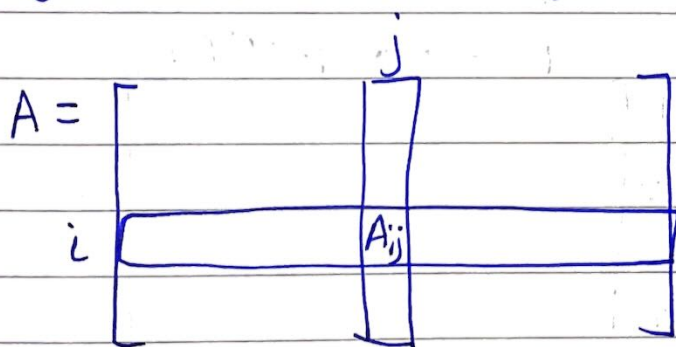
But the score $\langle h, a \rangle$ might go to ∞ . So we normalize it so that it doesn't become so large.

And then the auth and hub scores would converge which gives the HITS score for each page.

V17: Singular Value Decomposition (SVD)

Graph $u_i \rightarrow u_j$ (directed graph edges)

Adjacency matrix of G .



For 1.78M large dataset. Adjacency matrix A would be $1.78M \times 1.78M$

$$A_{ij} = \begin{cases} 1 & \text{if } u_i \rightarrow u_j \\ 0 & \text{otherwise} \end{cases}$$

A would be large and sparse matrix

$$A = U \Sigma V^T \text{ or } U S V^T$$

A is a $1.78M \times 1.78M$ binary sparse matrix.

u : $1.78M \times 6$ left singular matrix

$$\Sigma_{\alpha\beta} : 6 \times 6$$

V^T is $6 \times 1.78M$ right singular matrix

$$U = 19.5 \text{ MJ} \quad \begin{matrix} 1 & 2^0 & 3 & 4^0 & 5 & 6 \\ \left[\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 6 & 7 & 8 & 9 & 10 \\ 6 & 7 & 8 & 9 & 10 & 11 \end{array} \right] \end{matrix}$$

$$V^T = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{bmatrix} \quad 1 \dots 1.78M$$

V18: Weight features

Research paper formula is Used.

V19: Modeling

Random forest works well with non linear data.

F1 Score is the performance metric.

YT notes :-

- 1) Link prediction:- A network is changing over time. Given a snapshot of a network at time t , predict edges added in the interval (t, t') .
- 2) Link completion (missing links identification). Given a network, infer links that are consistent with structure, but missing (find unobserved edges).
- 3) Link reliability: Estimate the reliability of given links in the graph.

Predictions:- link existence (0/1) | link type
link weight (regression)

$$\text{No. of missing edges} = \frac{|V|(|V|-1)}{2} - |E|$$

In sparse graph $|E| \ll |V|^2$.

Prob. of correct random guess
 $O\left(\frac{1}{|V|^2}\right)$

Scoring algos:

Link prediction by proximity scoring

- ↳ For each pair of nodes compute proximity (similarity) score $c(v_i, v_j)$
- ↳ Sort all pairs by the decreasing score.
- ↳ Select top n pairs (or above some threshold) as new links.

Scoring functions

Local neighbourhood of v_i and v_j

- No. of common neighbours.

$$|N(v_i) \cap N(v_j)|$$

- Jaccard's Coefficient

$$\frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|}$$

- Adamic/Adar Index

$$\sum_{v \in N(v_i) \cap N(v_j)} \frac{1}{\log |N(v)|}$$

~~Shortest~~

paths and ensembles of paths
b/w v_i and v_j

- Shortest path: $-\min_s \{ \text{path}_{ij}^s > 0 \}$

- Katz Score

$$= \sum_{i=1}^{\infty} \beta^i |\text{paths}^{(i)}(v_i, v_j)|$$

$$= \sum_{i=1}^{\infty} (\beta A)_{ij}^i$$

$$= (I - \beta A)^{-1} - I$$

- Personalized (rooted) PageRank:

$$PR = \alpha (D^{-1}A)^T PR + (1-\alpha) |$$

- Expected number of random walk steps:

→ hitting time: $-H_{ij}$

→ Commute time $-(H_{ij} + H_{ji})$

→ Normalized hitting/commute time
 $(H_{ij} \pi_j + H_{ji} \pi_i)$

- SimRank

$$\text{SimRank}(v_i, v_j) = \frac{C}{|N(v_i)| \cdot |N(v_j)|} *$$

$$\sum_{m \in N(v_i)} \sum_{n \in N(v_j)} \text{SimRank}(m, n)$$

Normalization.

Vertex Feature Aggregations

- Preferential attachment

$$k_i \cdot k_j = |N(v_i)| \cdot |N(v_j)|$$

or

$$k_i + k_j = |N(v_i)| + |N(v_j)|$$

- Clustering Coefficient

$$cc(v_i) \cdot cc(v_j)$$

or

$$cc(v_i) + cc(v_j)$$

Research Preferential attachment :- paper

References:-

[1] Link prediction problem.

[4] Top most links of decreasing order of similarity score are likely to exist.

[8] AUC metric for link prediction.

Uses of Link prediction:-

- 1) link prediction can be used to find interactions between proteins in bioinformatics.

Ref: Airolidi et al. 2006

- 2) In ecommerce: Recommendation systems "people who bought this also bought" feature on Amazon.

Ref: Huang et al. 2005

- 3) Security domain: Identify hidden groups of terrorists or criminals

Ref: Hasan et al.

Problem def'n

Undirected, unweighted graph $G = (V, E)$

Given network $G[t_0, t'_0]$; output a list of edges not present in $G[t_0, t'_0]$ that are predicted to appear in the network $G[t_1, t'_1]$

$[t_0, t'_0]$ — Training Interval.

$[t_1, t'_1]$ — Test Interval.

S - Similarity matrix
 S_{xy} - Score b/w x and y

x & y belong to V (Set of vertices)

Non existent links are sorted in decreasing order of their scores

→ Top most links are more likely to exist.

Ref. [4]

observed links — E (90% of whole)
 ↓
 Training set (E_T)

E_P (remaining 10%) probe set.

$$E = E_T \cup E_P$$

$$\phi = E_T \cap E_P$$

Prediction performance metric = AUC

[8] Ref:

For n independent comparisons

n' = occurrences of missing links having a higher score

n'' = occurrences of missing links and nonexistent links having same score

$$AUC = \frac{n' + 0.5n''}{n}$$

Social Networks have power law degree distribution

Ref: Barabasi and Albert 1999 and

Small world Phenomenon

Ref: Watts and Strogatz, 1998 and

Community structure (Clustering effect)
Ref: Girvan and Newman 2002.

→ Nodes with high degrees are called hubs (hinge nodes).

→ Scale-free network: Most nodes links are very small in network. only a few nodes have lots of links.

Node - degree distribution is seriously uneven in large scale network.

(areto, heavy tailed or Zipfian degree distributions).

↑
WWW distribution

Small world Effect:

Average distance in the network is very small compared to the size of the network.

Each pair of nodes can be connected through a short path in a network.

Clustering Effect:

There is a circle of friends, acquaintances, rings and other small groups in social network.

Each member of the ^{small} group knows each other.

Also described by Triadic closure which is, there is many fully

connected subgraphs in social network.

Graph distance efficiency
Ref: [10].

Common Neighbours [7] Ref.

Preferential attachment:

$$|N(x)| \cdot |N(y)|$$